



DIGITAL WALL
FOR SYSTEM & SOFTWARE SERVICES LLC.

Sentry Siren USA.

KAZ - CCR

Preventive Maintenance Report

Project Name	Sentry Siren USA & Radio Control System Maintenance Contract
Client / Beneficiary	Basra Gas Company (BGC)
Contractor	Digital Wall for Systems and Software Service and General Trading LLC
Type Of Report	Preventive Maintenance Report
Facility Location :	KAZ – Khour Al Zubir
Siren System Location/Zone :	CCR ☒
Siren ID Number	KAZ-CCR-16V1T-B
Report Reference Number	PM-SRV-2026-001
Report Date	14-Apr-26
Prepared by	Hussein Ahmed Fadhil



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1. Introduction & Scope of Work

This document outlines the mandatory Preventive Maintenance (PM) procedures executed by Digital Wall engineering team for the Sentry Siren USA System located at the **KAZ – Khor Al-Zubair** facility.

On April 14, 2026, the designated team, led by Engineer Aws Watheq Nateq and supported by technicians Aref Mohammed and Abd Ali Waleed, arrived at the CCR to conduct a comprehensive system health check. The primary objective of this mobilization is to ensure 100% siren readiness for emergency notification protocols.

2. Site & System Identification

Field Details	Information / Data
Facility	KAZ – Khor Al Zubir
Siren Location / Area:	CCR ☒
Siren Model:	☒ 16V1T-B ☐ 3V8-H-B
Controller Type:	Gen 3 AC/DC Wireless
Siren ID / Number:	KAZ-CCR-16V1T-B
Engineers :	Aws Watheq Nateq
Technicians :	- Aref Mohammed - Abd Ali Waleed
Date Of Inspection :	Tuesday 14-4-2026

3. Siren Head & Mechanical Inspection

Task Description	Status	Comments / Actions Taken
Visual Inspection: Check siren head for bird nests, debris, or physical damage.	☒	The openings are clear of bird nests and trash. There are no signs of rust, cracks, or damage to the body of the unit.
Mounting Hardware: Ensure all bolts on hanging brackets and siren head are tight.	☒	All bolts for the bracket and siren head are tight, but they are starting to show some surface rust.
Audible/Test Run: Perform a 3-second "Growl Test" or full activation via Gen 3 Controller.	☐	No Audible / Test Performed

4. GEN 3 Wireless Radio Controller

Task Description	Status	Technical Reading
Power Supply : Verify 220VAC to 24VDC output.	☒	The system is getting a steady 220V AC from the main power and putting out a solid 24.09V DC. Everything is stable at this point.

Radio Signal : Check Maxon SD-171EX Radio signal strength/reception.	☒	Signal within operational parameters
Relay Check : Inspect Omron LY2-D activation relays for carbon/wear.	☒	Active and no carbon / wear
Converter : Test Switch Mode DC-DC Converter (AC Failure Simulation).	☒	13.4VDC Output (Good) 24VDC input (Good)
Antenna : Inspect VHF Antenna connections for corrosion or loose cables.	☒	All antenna connections and cabling are verified tight, secure, and completely free of corrosion.
Decoder Board : Check All PCB Health Check & wiring	☒	- Decoder Board is Good. - The current wire connections non-compliance and don't look professional because they are missing ferrules (wire-end sleeves).
Starter : Check 24V to 12V Converter output in Dual DC starter.	☒	Input 24VDC/Output 13.45VDC (Good)

5. Battery & Charging System (Optima 34R RED TOP)

Task Description	Status	Technical Reading
Battery Health : Inspect (6) 34R batteries for swelling, leaks, or terminal corrosion.	☒	- Battery Health is Good, no leaks - The main neutral wire is missing its terminal connector. Also, the terminals on Battery 1 and Battery 5 have some rust and buildup (corrosion) on them
Voltage Check : Measure individual battery voltage (Target: 12V+).	☒	Batt 1: 12.82VDC (Good) Batt 2: 12.71 VDC (Good) Batt 3: 12.97 VDC (Good) Batt 4: 12.75 VDC (Good) Batt 5: 12.71 VDC (Good) Batt 6: 12.68 VDC (Good)
NOCO Charger : Inspect LED status (Solid Green = Charged).	☒	All solid green
Cleanliness : Clean battery terminals and apply anti-corrosion grease.	☒	All battery terminals need anti corrosion grease , Cleaning
Battery Center : Battery Centre Box with Batteries Package Full Inspection	☒	The box needs to reseal/replace because water and dust can leak inside.

6. Power Distribution & Starter

Task Description	Status	Comments
Rectifier : Inspect Transformer Rectifier (16V1T-B) for heat/hum.	☒	No heat or humidity found (good)
Blocks : Ensure Power Distribution Blocks (Marathon) are secure.	☒	Secure and working perfect (good)
Relay : Verify Eaton D3PF2AB Relay operation. , and other related relays 220 Relay	☒	Working Perfect (good)

7. Portable Units

Task Description	Status	Comments
VHF Radio : Function test handheld & verify battery charge.	☒	Working Perfect (good)

Additional Comments & Report Summary :

- 1- DC-DC Converter: The converter is loose and not mounted properly to the board. We need to secure it and redo the input and output wiring connections to make sure they stay tight and don't vibrate loose.
- 2- Controller Wiring: The wires on the Gen 3 Wireless Radio Controller/Decoder circuit board need to be re-connected correctly to prevent any future loose connections.
- 3- Wiring Standards: The current wiring looks unprofessional and doesn't meet standards because the wire ends are missing ferrules (protective metal sleeves). These need to be added for a secure, professional fit.
- 4- Neutral Connection: The main neutral wire is missing its terminal connector, which is a key part of the electrical setup that needs to be fixed.
- 5- Battery Maintenance: All battery terminals and tops need to be thoroughly cleaned. Once the rust and buildup (corrosion) are removed—specifically on Battery 1 and Battery 5—we need to apply anti-corrosion grease and tighten them to the correct specs.
- 6- Battery Enclosure: The battery center box is not sealed properly, allowing dust and water to leak inside. It needs to be resealed or replaced to protect the Optima 34R Red Top batteries.
- 7- Transformer Rectifier (Critical): The Transformer Rectifier (Model 16V1T-B) is outputting 120 VDC, which is 20V higher than the manufacturer allows. This unit is failing and needs to be replaced immediately to avoid damaging the rest of the system.
- 8- System Grounding: During our survey, we found that neither the electrical system nor the metal structure of the siren is grounded. This is a safety risk that needs to be corrected.

9. Contractor Certification & Client Acceptance

Contractor Statement: We certify that the equipment listed above has been inspected and tested according to the manufacturer's specifications. The system is currently in [**Optimal / Restricted / Offline**] status.

Digital Wall representative	BGC – Basra Gas Company representative
Sign: 	Sign: 
Name: Aws watheg	Name: Husam Jabbar
Title: Eng	Title: E.Ng
Date: 14-4-2026	Date: 14-4-2026



شركة السور الرقمي لخدمات النظم والبرمجيات
والتجارة العامة محدودة المسؤولية
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